

ACADEMIC TRACK: APPLIED CRYPTOGRAPHY & SECURITY ENGINEERING

Applied Cryptography: Memory-Hard Key Derivation & Authenticated Encryption

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Abstract: Password storage and session token derivation demand modern memory-hard cryptographic functions to resist GPU and ASIC-accelerated brute-force attacks. This report provides a comparative security analysis between Argon2id, PBKDF2, and bcrypt within the context of academic credential protection. We further detail the integration of authenticated AES-256-GCM symmetric encryption for student artifact confidentiality and verify constant-time HMAC comparison routines.

1. Memory-Hard Key Derivation (Argon2id vs PBKDF2)

Argon2id combines data-independent and data-dependent memory access patterns, providing superior defense against both cache-timing attacks and hardware-parallelized cracking rigs. We benchmarked Argon2id with memory cost $m=65536$ KiB, time cost $t=3$ iterations, and parallelism $p=4$ threads against legacy PBKDF2 with 310,000 iterations.

2. Symmetric Cipher Security (AES-256-GCM)

For artifact persistence at rest, AES-256 in Galois/Counter Mode (GCM) guarantees both confidentiality and authenticity. Each file is paired with an ephemeral 96-bit initialization vector (IV) generated via a CSPRNG. The 128-bit authentication tag ensures immediate detection of bit-flipping tampering.

Algorithm / Cipher	Target Function	Memory Cost	Iterations	Crack Cost (\$/MH)	Standard
Argon2id	Password Hashing	64 MB	$t = 3$	> \$500,000	RFC 9106 / NIST
bcrypt	Legacy Password	4 KB	cost = 12	> \$45,000	OpenBSD Spec
PBKDF2-HMAC	Key Derivation	negligible	310,000	< \$2,500	NIST SP 800-132
AES-256-GCM	Payload Encryption	negligible	Single-pass	Infeasible	FIPS 197 / NIST

3. Constant-Time HMAC Verification

To prevent timing side-channel attacks during authentication token and audit ledger verification, string comparisons utilize `MessageDigest.isEqual()` and `hash_equals()`, guaranteeing constant execution time regardless of matching prefix length.

4. Laboratory Conclusions

The cryptographic architecture adheres to NIST SP 800-63B standards and guarantees verifiable defense-in-depth across all storage and transport boundaries.